REAL-TIME-VIDEO-ANALYSIS-USING-YOLOV9-AND-OPENCV

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ABSTRACT

This project focuses on Image and Video Analysis using YOLO for real-time object detection. YOLO (You Only Look Once) is a state-of-the-art deep learning algorithm that detects and classifies multiple objects in a single pass, making it highly accurate and efficient. The system takes images or videos as input and identifies objects by drawing bounding boxes and displaying their corresponding class labels with confidence scores. YOLO provides fast processing speed while maintaining high detection accuracy, making it suitable for real-time applications.

This project demonstrates the capability of YOLO to analyze visual data effectively and can be applied in areas such as surveillance, traffic monitoring, autonomous vehicles, and smart security systems. The results show that YOLO is a powerful and reliable solution for image and video analysis through real-time object detection.

Experiment: Real-Time Road Traffic Object Detection using YOLOv9

CODE :

```
# Install YOLO
!pip install ultralytics -q
# Run detection
!yolo detect predict model=yolov9m.pt source="/content/road_trafifc.mp4"
# Convert output video to mp4
!ffmpeg -i runs/detect/predict/road_trafifc.avi -vcodec libx264 output.mp4
# Download
from google.colab import files
files.download("output.mp4")
```

OUTPUT:

Ultralytics 8.4.104 🚀 Python-3.12.13 torch-2.11.0+cpu CPU (Intel Xeon CPU @ 2.20GHz)
YOLOv9m summary (fused): 151 layers, 20,070,832 parameters, 0 gradients, 76.8 GFLOPs
video 1/1 (frame 1/301) /content/road_trafifc.mp4: 384x640 4 persons, 7 cars, 2 motorcycles, 1 bus, 1 traffic light, 1 bench, 1 backpack, 1068.3ms
video 1/1 (frame 2/301) /content/road_trafifc.mp4: 384x640 3 persons, 6 cars, 2 motorcycles, 1 bus, 1 traffic light, 1 bench, 1 backpack, 915.5ms
video 1/1 (frame 300/301) /content/road_trafifc.mp4: 384x640 3 persons, 10 cars, 2 motorcycles, 1 bus, 894.5ms
video 1/1 (frame 301/301) /content/road_trafifc.mp4: 384x640 3 persons, 10 cars, 2 motorcycles, 1 bus, 916.8ms
Speed: 1.7ms preprocess, 994.6ms inference, 1.3ms postprocess per image at shape (1, 3, 384, 640)
Results saved to **/content/runs/detect/predict**
💡 Learn more at https://docs.ultralytics.com/modes/predict

RESULTS :

INPUT : Actual video (frame)

OUTPUT : After applying YOLO algorithm

As seen in the output frame, YOLO successfully detects and labels multiple road-traffic objects including persons, cars, motorcycles, a bus, and a traffic light, each with a bounding box and an associated confidence score, confirming the model's effectiveness for real-time traffic-scene analysis.